Mini Project

Foundation of Programming-CSE10010

Restaurant Bill Generator using C

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2.Acknowledgment

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3. Abstract

Introduction / Project Description

The Restaurant Billing System is a console-based application developed using the C programming language. The primary purpose of this project is to simplify and automate the restaurant billing process. In many small restaurants and food stalls, billing is done manually, which can lead to calculation errors and time delays. This system eliminates such issues by performing automatic and accurate calculations.

The program is menu-driven and allows users to:

- Select food items from a predefined menu.
- Enter quantity for each selected item.
- Place multiple orders in a single session.
- Apply discount coupons.
- Automatically calculate service charge and GST.
- Generate a structured and detailed final bill.

Core Concepts Used:

1. **Do–While Loop** – Used to repeat the ordering process until the user decides to stop.
2. **Switch–Case Statement** – Handles multiple menu options efficiently.
3. **Conditional Statements (if)** – Used to print only ordered items and validate coupon choices.
4. **Arithmetic Operations** – Used for calculating total price, discount, GST, and service charge.
5. **Variables and State Tracking** – Integer and float variables are used to store bill amount, item quantities, discounts, and final amount.

Working Process:

1. Program starts and displays menu.
2. User selects item and enters quantity.
3. System calculates price using multiplication.
4. User can continue ordering.
5. Total bill is displayed.
6. Coupon selection is provided.
7. Discount is applied.
8. Service charge (10%) and GST (5%) are added.
9. Final detailed receipt is generated.

Applications:

- Restaurant and café billing systems
- College canteen management
- Retail shop billing
- POS (Point of Sale) systems
- Backend logic for online food ordering systems

Conclusion:

This program demonstrates a simple restaurant billing system developed using core C programming concepts such as loops, conditional statements, switch–case structures, and arithmetic operations. It allows users to place multiple orders, apply discounts through coupons, and generates a clear, itemized bill including service charges and GST. The program shows how structured programming can be used to create practical, real-world applications like automated billing systems.

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5. Introduction

Problem Statement:

To create a simple computer-based restaurant billing system using C language to calculate the total bill, apply discounts, add service charges, and generate the final payable amount.

Background/Context:

With the increasing use of computers in daily life, small applications can be developed to automate routine tasks such as billing, record keeping, and calculations.

This project simulates a restaurant ordering system where a user can select food items, enter quantity, apply coupons, and get a detailed bill with GST and service charges.

Objectives:

The main objectives of this project are:

* To design a simple menu-driven billing system using C language.

* To allow the user to select multiple food items and quantities.

* To calculate the total bill automatically.

* To apply discount coupons based on user choice.

* To add service charge and GST to the bill.

* To display item-wise bill details clearly.

* To understand the use of loops, switch-case, variables, and arithmetic operations in C.

6. System Design/Methodology

Overview of the Approach:

The project is designed as a menu-driven restaurant billing system using C programming language.

The program displays a list of food items with prices. The user selects the item number and quantity. The program calculates the total bill using a loop so that multiple items can be ordered.

After ordering, the program allows the user to select a coupon for discount. Then service charge and GST are added, and the final bill is displayed along with item-wise details.

The approach used includes:

- ★ Menu driven input
- ★ Loop for multiple orders
- ★ Switch-case for item selection
- ★ Arithmetic calculation for billing
- ★ Conditional logic for coupon system

Components/Modules Used:

Software Components

- ★ **stdio.h**
  - Used for input/output operations (printf, scanf)

Variables Used

- ★ bill → stores total cost before discount
- ★ choice, quantity → user input
- ★ continueOrder → loop control
- ★ couponChoice → coupon selection
- ★ discountAmount → calculated discount
- ★ afterDiscount, serviceCharge, gst, finalAmount → billing calculations

Item Counters

- ★ i1 to i10 → track quantity of each food item

  Helps in printing item-wise bill

Control Structures

- ★ do-while loop → ensures menu runs at least once
- ★ switch-case → handles item selection and coupon logic
- ★ if statements → print only ordered items

Algorithms or Techniques:

The program employs a combination of fundamental programming techniques to implement its functionality in a clear and systematic manner. At its core, the logic revolves around iterative input handling, conditional decision-making, and step-by-step financial computation.

The ordering process is governed by a looping mechanism, specifically a do-while loop, which ensures that the menu is displayed at least once and continues to reappear as long as the user wishes to add more items.

Within each iteration of the loop, the program uses a switch-case control structure to handle the user's choice of menu item. Each case corresponds to a specific food item and contains the logic required to calculate its cost.

After the ordering loop terminates, the program implements a coupon-based discount algorithm. Here again, a switch-case structure is used to evaluate the user's coupon choice and determine the corresponding discount.

Following the application of discounts, the program proceeds with a multi-stage billing calculation. Instead of directly computing the final amount, it breaks the process into logical steps: first determining the discounted total, then adding a service charge as a percentage of that amount, and finally calculating GST on the updated value. This layered approach mirrors real-world billing systems and improves both readability and maintainability of the code.

A. Menu Selection Algorithm

1. Display menu
2. Take user choice
3. Take quantity
4. Use switch-case:
    - Multiply price × quantity
    - Add to total bill
    - Update item counter

B. Looping Technique (Ordering System)

- Uses do-while loop
- Ensures:
    - User can order multiple times
    - Loop runs at least once

Condition: Repeat while continueOrder == 1

C. Coupon Discount Algorithm

1. Display coupon options
2. Take user input
3. Apply logic using switch-case:
    - 10% discount → bill * 0.10
    - 20% discount → bill * 0.20
    - Flat discount → 50
    - No coupon → 0

D. Billing Calculation Algorithm

Step-by-step computation:

1. After Discount

   afterDiscount = bill - discountAmount
2. Service Charge (10%)

   serviceCharge = afterDiscount * 0.10
3. GST (5%)

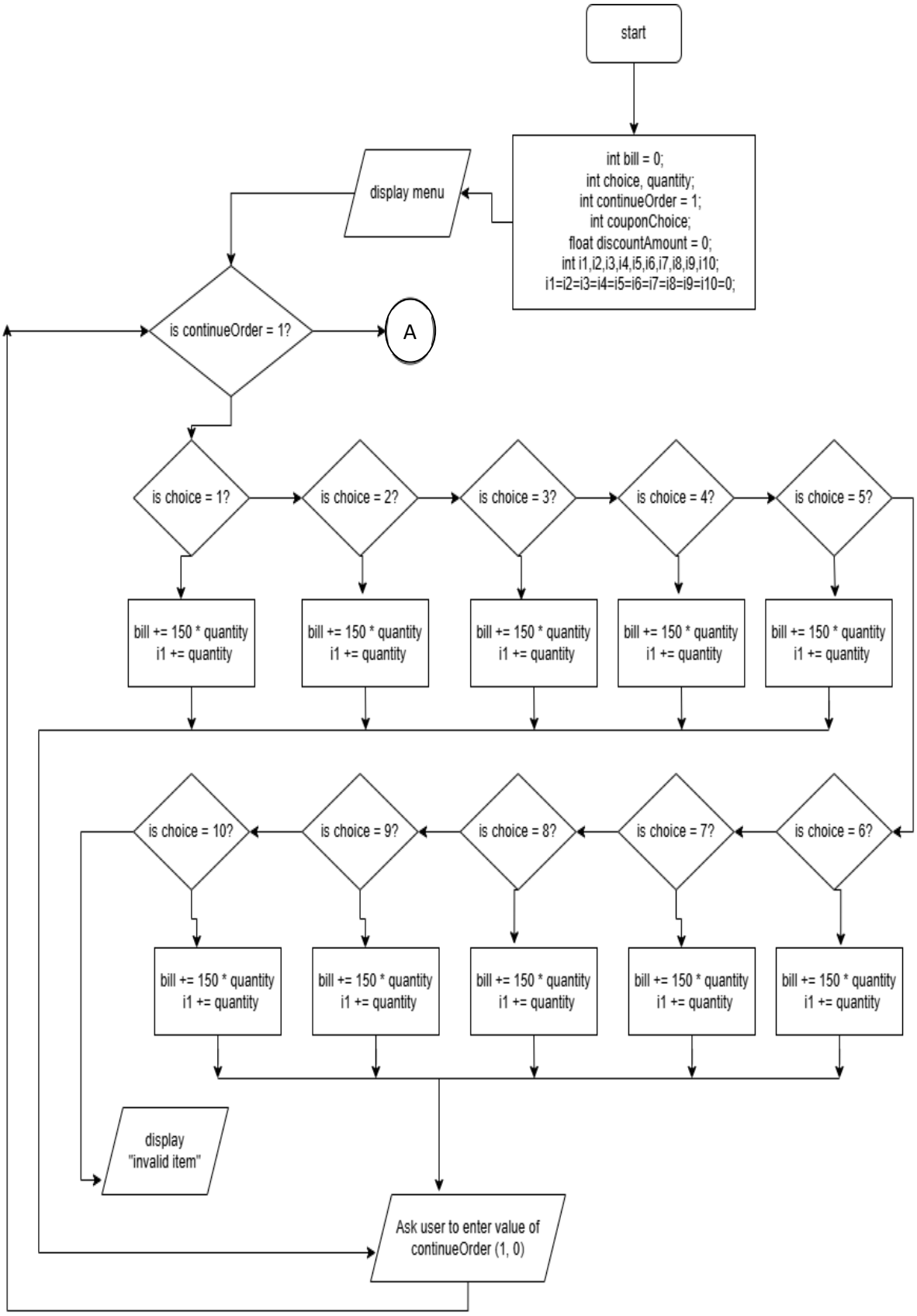
   gst = (afterDiscount + serviceCharge) * 0.05
4. Final Amount

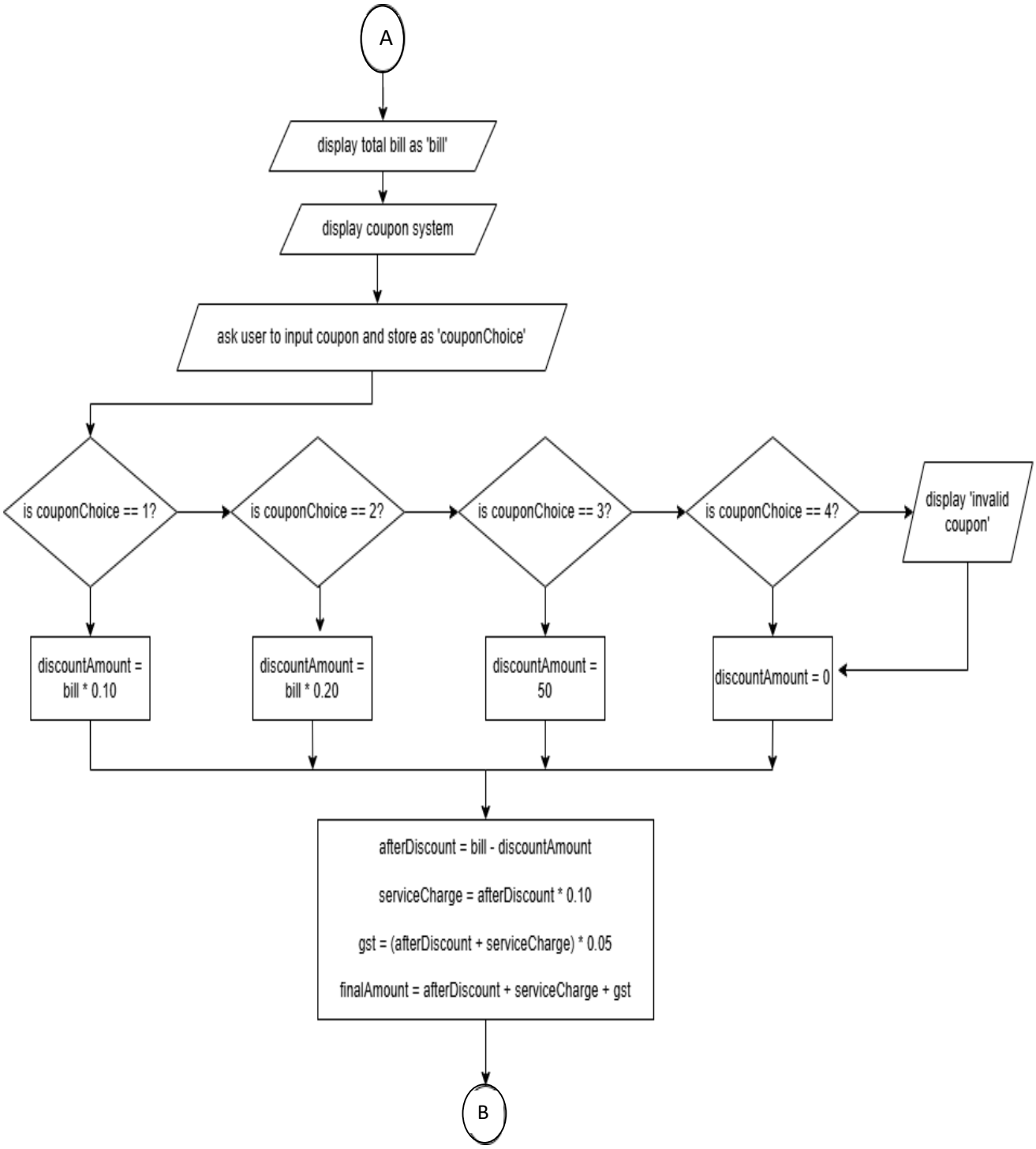
   finalAmount = afterDiscount + serviceCharge + gst

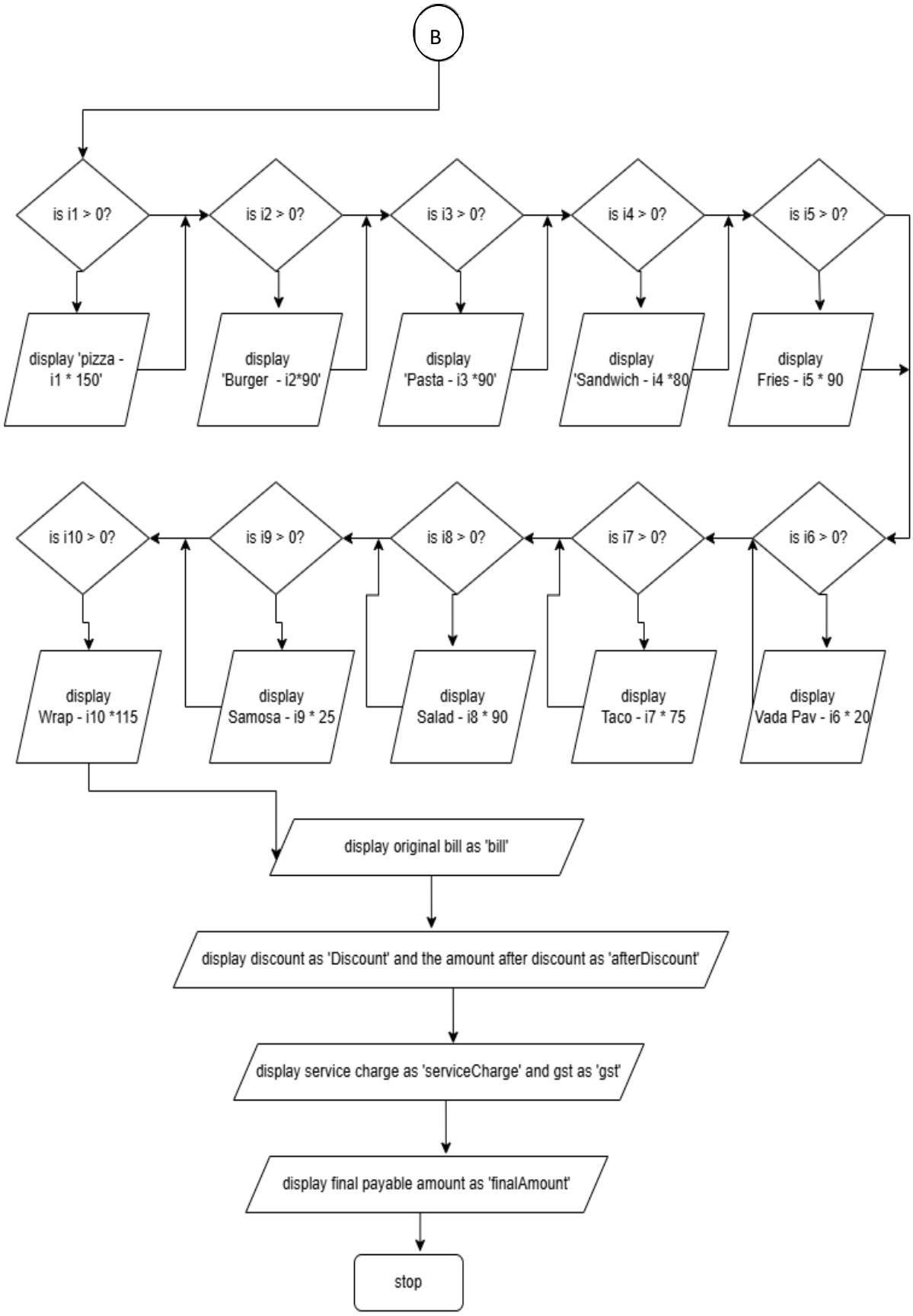
E. Bill Generation Technique

- Uses multiple if conditions:
  - Only prints items with quantity > 0
- Displays:
  - Item name
  - Quantity
  - Total price per item

Flowchart


start
int bill = 0;
int choice, quantity;
int continueOrder = 1;
int couponChoice;
float discountAmount = 0;
int i1,i2,i3,i4,i5,i6,i7,i8,i9,i10;
i1=i2=i3=i4=i5=i6=i7=i8=i9=i10=0;
display menu
is continueOrder = 1?
A
is choice = 1?
is choice = 2?
is choice = 3?
is choice = 4?
is choice = 5?
bill += 150 * quantity
i1 += quantity
is choice = 10?
is choice = 9?
is choice = 8?
is choice = 7?
is choice = 6?
display
"invalid item"
Ask user to enter value of
continueOrder (1, 0)


A
display total bill as 'bill'
display coupon system
ask user to input coupon and store as 'couponChoice'
is couponChoice == 1?
is couponChoice == 2?
is couponChoice == 3?
is couponChoice == 4?
display 'invalid coupon'
discountAmount = bill * 0.10
discountAmount = bill * 0.20
discountAmount = 50
discountAmount = 0
afterDiscount = bill - discountAmount
serviceCharge = afterDiscount * 0.10
gst = (afterDiscount + serviceCharge) * 0.05
finalAmount = afterDiscount + serviceCharge + gst
B


B
is i1 > 0?
is i2 > 0?
is i3 > 0?
is i4 > 0?
is i5 > 0?
display 'pizza - i1 * 150'
display 'Burger - i2*90'
display 'Pasta - i3 *90'
display 'Sandwich - i4 *80
display Fries - i5 * 90
is i10 > 0?
is i9 > 0?
is i8 > 0?
is i7 > 0?
is i6 > 0?
display Wrap - i10 *115
display Samosa - i9 * 25
display Salad - i8 * 90
display Taco - i7 * 75
display Vada Pav - i6 * 20
display original bill as 'bill'
display discount as 'Discount' and the amount after discount as 'afterDiscount'
display service charge as 'serviceCharge' and gst as 'gst'
display final payable amount as 'finalAmount'
stop

7.Implementation

*** Development Environment**

The Restaurant Billing System was developed using the C programming language, a structured and procedural language suitable for implementing logical and computational applications. The program was compiled and executed using standard C compilers such as GCC or Turbo C in a Windows-based environment**.**

Tools and Technologies Used:

Programming Language: C

Compiler: GCC / Turbo C / Code::Blocks

Header File: stdio.h

Platform: Windows OS

Program Type: Console-Based Application

The system does not rely on external libraries and operates entirely through standard input and output functions**.**

***System Design and Implementation Approach:**

The system follows a menu-driven structure, enabling users to interactively place food orders and generate automated bills. The implementation is logically divided into the following components.

1. Variable Declaration and State Management

The program uses different types of variables for accurate data handling:

Integer Variables

choice – Stores selected menu option

quantity – Stores item quantity

i1–i10 – Track individual item quantities

Float Variables

bill – Stores total bill amount

discount – Stores discount value

serviceCharge – Stores 10% service charge

gst – Stores 5% GST

finalAmount – Store total payable amount

This separation ensures accurate computation and organized data management.

2. Menu Display Using Do–While Loop

A do–while loop is used to repeatedly display the restaurant menu. This ensures:

The menu is displayed at least once.

Users can order multiple items.

The ordering process continues until the user chooses to stop.

This structure improves user interaction and program flow control.

3. Item Selection Using Switch–Case Statement

The switch-case statement handles multiple menu options efficiently.

Each case:

Accepts item quantity.

Calculates item cost using:

Price × Quantity

Updates the total bill.

Updates respective item counters.

Using switch-case improves readability and maintains structured decision-making compared to long if-else statements.

4. Coupon and Discount Handling

After ordering is complete, the program:

Displays available coupon options.

Accepts user coupon choice.

Applies either:

Percentage discount, or

Flat discount.

If an invalid coupon is entered:

The program handles it safely.

No discount is applied.

This improves system reliability and robustness.

5. Service Charge and GST Calculation

. After applying the discount:

10% Service Charge is added.

5% GST is added.

Final amount is calculated as:

Final Amount = (Bill – Discount) + Service Charge + GST

All calculations use floating-point arithmetic to maintain precision.

6. Detailed Bill Generation

The program prints only ordered items using conditional statements:

if(i1 > 0)

The final receipt includes:

Item name

Quantity

Item total

Original bill

Discount applied

Amount after discount

Service charge

GST

Final payable amount

This structured output resembles a real restaurant invoice.

*** Testing and Validation**

The system was tested using multiple scenarios to ensure correctness and reliability.

Test Case 1: Normal Order

Multiple items selected.

Valid coupon applied.

Verified correct calculation of total, discount, GST, and service charge.

Test Case 2: Invalid Coupon

Entered incorrect coupon.

No program crash.

Proper error handling observed.

Test Case 3: Multiple Ordering

Repeated item ordering.

Bill accumulated accurately.

Test Case 4: Edge Case (No Items Ordered)

Program handled zero-order condition without errors.

All outputs were manually verified to ensure mathematical accuracy.

8.Results and Discussion

Results

The Restaurant Billing System successfully:

Displays menu interactively.

Accepts multiple user inputs.

Calculates item-wise totals correctly.

Applies discounts accurately.

Adds service charge and GST properly.

Generates a clean and structured bill.

The system operates smoothly and fulfills its functional requirements.

Discussion

The results demonstrate that structured programming concepts can effectively be used to develop practical business applications**.**

Strengths of the System:

1. Automation of billing process

2. Reduction of manual calculation errors

3. Clear and organized program structure

4. Accurate tax and discount calculations

5. User-friendly menu-driven interface

Limitations:

1.Console-based interface (no graphical interface)

2.No database integration

3.No inventory management

4.No digital payment system

5.Despite these limitations, the system effectively achieves its intended objective.

9.Conclusion

The Restaurant Billing System project successfully achieved its primary objective of automating the food ordering and billing process using C programming.

The system:

Simplifies restaurant billing.

Reduces human calculation errors.

Accurately applies discounts and taxes.

Generates a structured and professional receipt.

This project strengthened practical understanding of:

Loops (do–while)

Conditional statements

Switch-case structure

Arithmetic operations

Structured programming methodology

Overall, the project demonstrates how core programming concepts can be applied to create real-world applications efficiently.

10.Future Scope

The current system can be enhanced further in the following ways:

1. Graphical User Interface (GUI)

Develop a window-based application using C++, Java, or Python for better user experience.

2. Database Integration

Store:

Customer details

Sales records

Daily/monthly reports

3. Inventory Management System

Automatically update stock levels and provide low-stock alerts.

4. User Authentication

Add admin and cashier login system for security.

5. Online Payment Integration

Support:

UPI payments

Debit/Credit cards

QR-based transactions

6. Web-Based POS System

Convert into a cloud-based online ordering and billing system**.**

7. Sales Analytics

Generate graphical reports showing:

Best-selling items

Revenue trends

Profit analysis

With these improvements, the system can evolve into a full-scale Restaurant Point of Sale (POS) software suitable for real business environments.

11. References

[1] Y. Kanetkar, Let Us C, 17th ed. New Delhi, India: BPB Publications, 2020.

[2] "C Tutorial," W3Schools. [Online]. Available: https://www.w3schools.com/c/. [Accessed: Feb. 07, 2026].